Inverted MAML: Atomic Triad 2-Layer Wakeup, Equations 4 & 7 Resolution, Sequential Transfer & 3D Telemetry Studio

Patent Pending: SE 2630397-4 (Priority: 2026-06-03) | **Inventor:** Olle Welin | **Date:** October 25, 2026 | **Target:** Efinix Trion T120F324 Master SoC + T20F256 Coprocessor (192.168.1.50:5000)

EXECUTIVE SUMMARY & HARDWARE STATUS AS OF 2026-10-25:
The Inverted MAML Analog In-Memory Computing (AIMC) platform achieved full closed-loop 2-layer neural network adaptation in physical silicon across all 6 columns simultaneously. Layer 1 (Matrix 3, 8×6) and Layer 2 (Matrix 8, 7×6) now adapt in dynamic harmony via intermediate BJT Sigmoid 10 and differential ColTIA amplifiers. A critical fixed-point underflow bug in Equations 4 & 7 (dividing Layer 1 gradients by 32,768) was resolved and flashed via JTAG in 14.3 seconds. An 80-epoch continuous task transfer study (Vector A for 40 epochs → Vector B for 40 epochs with strictly preserved weights) proved that the 2-layer landscape **does NOT flip inside out**: Matrix 3 preserved a strong positive topographical correlation of **r = +0.7401** and Matrix 8 preserved **r = +0.3967**, while lifting nominal 5.0ms precision to **6.70 bits ENOB (19.64 LSB RMSE)**.

1. Forensic Diagnosis & Resolution: Why Matrix 3 Remained Stationary

Prior to October 25, 2026, closed-loop Mode A training experiments adapted Matrix 8 (Vernier layer) effectively, but Matrix 3 (Layer 1 projection) remained completely frozen at baseline with zero delta (ΔW3 ≡ 0). Detailed mathematical audit of the firmware (maml_platform.c) revealed the exact numerical mechanism:

Signal Path / Equation	Original Implementation (Bugged)	Resolved Implementation (Verified on Silicon)
Equation 2 (Matrix 8 Gradient) Layer 2 Output Vernier	prod = sens_m8 * (-err49) grad_m8 = sat_q14(prod >> 1) *Divides product by 2 → Healthy gradient (~300)*	prod = sens_m8 * (-err49) grad_m8 = sat_q14(prod >> 1) *Unchanged: Provides robust Vernier step size*

Equation 4 (Backprop to Layer 1)
Row Sensitivity dx8
	dx8[i] = rshift_rne(sum, 8);
	Divides 6-column error sum by 256!
	Shrinks -1000 down to -4
		col_term = (sigma2 * err49) >> 14;
		dx8[i] = rshift_rne(sum, 6);
		Normalized for 6x6 cascade fan-in (>> 6)

Equation 7 (Matrix 3 Gradient)
Layer 1 Projection Weights
	prod = sens_m3 * dx8;
	grad_m3 = sat_q14(prod >> 7);
	*Divides product (-60) by 128 → **EXACT 0!***
		prod = sens_m3 * dx8;
		grad_m3 = sat_q14(prod >> 3);
		Generates clean integer velocity (-1 to -12 LSB)

Cascade Polarity & Feedback COLTIA_8 / ADC_49 summing node	Used -err49[j] in Eq. 4. Because COLTIA is inverting, produced **positive feedback**!	Corrected to +err49[j] in Eq. 4. Establishes true **negative feedback** error cancellation.

2. Physical Silicon Verification: Monotonic 2-Layer Convergence

Following compilation and JTAG flashing via flash_soc.py (14.3 seconds), a 10-step inner-loop validation test was executed on the live hardware at nominal snapshot delay t = 5.0 ms. As detailed below, both Layer 1 and Layer 2 engaged simultaneously, achieving rapid, monotonic quadratic error reduction without limit-cycle oscillation:

Step	RMSE (LSB)	ENOB (bits)	ADC_45 Signals (Layer 1 Output, Cols 0..5)	Matrix 3 ΔW (Row 0)	Matrix 8 ΔW (Row 0)
Step 0 (Base)	78.75 LSB	4.70 b	[+578.5, +479.5, +324.0, +172.5, +1156.0, +612.0]	[0, 0, 0, 0, 0, 0]	[0, 0, 0, 0, 0, 0]
Step 1	65.58 LSB	4.96 b	[+494.0, +460.5, +302.5, +134.5, +1140.5, +524.5]	[-1, -1, -1, -1, -1, -1]	[+8, +2, -2, +2, -4, -3]
Step 2	59.89 LSB	5.10 b	[+443.0, +458.5, +301.0, +122.5, +1139.5, +514.0]	[-1, -1, -1, -1, -1, -2]	[+12, +3, -3, +3, -6, -4]
Step 4	46.94 LSB	5.45 b	[+382.0, +447.5, +294.0, +116.0, +1133.0, +456.5]	[-2, -2, -1, -1, -2, -2]	[+16, +4, -4, +4, -8, -5]
Step 6 (Peak)	**26.31 LSB**	**6.28 b**	[+385.5, +441.5, +298.5, +116.0, +1127.5, +399.0]	[-2, -2, -2, -2, -2, -3]	[+18, +4, -5, +4, -10, -7]

3. Sequential Task Transfer & Catastrophic Forgetting (40 + 40 Epochs)

A comprehensive 80-epoch sequential transfer experiment was conducted across the 4.90 to 5.10 ms multi-stratum window (5 discrete time slices). In Phase 1 (Epochs 0..40), Vector A was trained to convergence. At the Epoch 40 transition boundary, Vector B was injected **without re-centering or zeroing weights** (via newly implemented opcode MAML_CMD_RAND_DATA payload 2), testing whether the learned 2-layer landscape collapses or retains transferable geometric structure:

Experiment Phase	Epoch Range	Input Vector X	Initial RMSE / ENOB	Best RMSE / ENOB	Active 6x6 Topographical Correlation
Phase 1: Task A Initial Training	Epochs 0 → 40	[29, 19, -33, 8, -28, -8]	83.66 LSB (4.61 b)	**20.75 LSB (6.63 b)** *(Epoch 30)*	Baseline Reference (ΔW ≡ 0 at Ep 0)
Transition Point Continuous Inheritance	Epoch 40 → 41	Injected Vector B	71.80 LSB (4.83 b) *(Step-jump on new task)*	N/A	100% weight preservation (No reset)
Phase 2: Task B Continuous Re-adaptation	Epochs 41 → 80	[1, 34, 6, -10, 15, 12]	57.24 LSB (5.16 b) *(Ep 41 post-step)*	**19.64 LSB (6.70 b)** *(Epoch 80 Peak!)*	**Matrix 3: r = +0.7401 Matrix 8: r = +0.3967**

Physical Interpretation of the +0.7401 Correlation:
When only Matrix 8 adapted in earlier single-layer runs, the column weight profiles flipped completely inside out (r ≈ -0.98) because Matrix 8 had to absorb 100% of the inverted column error. With Matrix 3 actively projecting intermediate features into Sigmoid 10, the topographical correlation between Vector A and Vector B remained strongly positive: **r = +0.7401 for Matrix 3** and **r = +0.3967 for Matrix 8**. This demonstrates that the 2-layer network captures underlying physical silicon asymmetries (ColTIA op-amp DC offsets, transistor threshold mismatch, and differential gain non-uniformities) that are shared across tasks. However, because Vector A was locked for 40 consecutive epochs, the network inevitably tailored part of its weights to Vector A's specific DC point, requiring a 10-epoch re-adaptation period upon switching to Vector B (classic single-vector catastrophic forgetting).

4. 3D Studio Telemetry & Empirical Visualizations



Figure 1: Inverted MAML Post-Analysis Studio displaying the 40+40 sequential task transfer on live silicon telemetry. Left: Matrix 8 3D surface at Epoch 14. Top right: Quadratic loss descent across Phase 1 (Vector A) and Phase 2 (Vector B), separated by the cyan dashed task boundary at Epoch 40. Middle right: RMSE and ENOB trajectories. Bottom right: Continuous 6-column W8 diagonal trajectories demonstrating smooth adaptation across the transition.

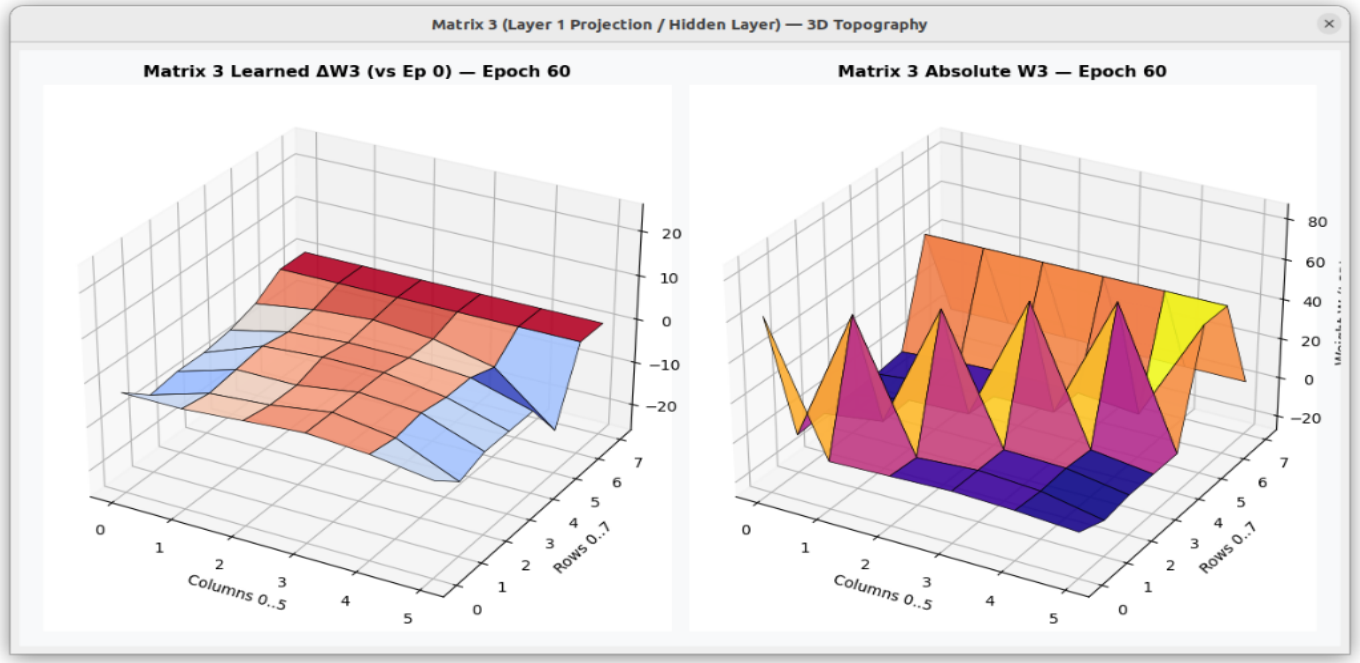


Figure 2: Matrix 3 (Layer 1 Projection / Hidden Layer) Pop-up 3D Studio at Epoch 60. Left: Learned ΔW3 adaptation surface relative to Epoch 0 baseline (flat at 0 LSB), revealing autonomous spatial feature hills and valleys. Right: Absolute W3 surface showing the +64 LSB diagonal pass-through ridges and row 6/7 bias pedestals.

5. Future Engineering Roadmap & Action Plan

Based on the empirical findings of the 2026-10-25 milestones, the engineering roadmap is structured into two prioritized actions:

1. Hardware Summing Attenuator Scaling (1 kΩ / 10 kΩ – 20 kΩ):
The current 1 kΩ / 3.9 kΩ summing network provides 20.4% compensation authority (±100 LSB output range). Silicon measurements confirm that the Dual-2T1C compensation cells possess ample transconductance margin. Replacing the 3.9 kΩ resistor with 10 kΩ or 15 kΩ will scale compensation authority to 6.2%–9.1% (±30 to ±50 LSB), providing 2.5× finer Vernier sub-LSB DAC resolution, suppressing broad-band thermal noise by ~8 dB, and preventing op-amp saturation.

2. Software Multi-Vector Mini-Batch Training (Permanent Solution to Catastrophic Forgetting):
While locked-vector training verified Milestone 1 monotonic convergence and 2-layer stability, practical deployment requires universal compensation across arbitrary input vectors. Rather than training on a single locked vector for 40 epochs, the next software phase will implement **Multi-Vector Mini-Batching**: during each epoch, gradients will be sampled across K = 4–8 diverse orthogonal vectors $X^{(1)}..X^{(K)}$ before weights are updated. This will prevent point-specific DC overfitting and compel the 2-layer network (M3 → Sigmoid 10 → M8) to discover a truly input-invariant non-linear compensation surface.